

VARUN SREENIVASAN

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EXPERIENCE

DataChat, Inc., Madison, WI

Feb 2022 - Present

Senior Software Engineer

- *Scalability Upgrades*: Proposed & implemented [Artillery](#) driven load tests, fixed bottlenecks, and boosted platform functionality.
- *Led a team of developers* to overhaul & integrate VectorDB platform with critical providers Google Vector Search & Milvus.
- *Increased Automation & Delivery Cadence*: Led CI/CD push to streamline release mgmt. process involving [Kubernetes](#) deployments.
- *Accelerated LLM adoption*: Set up evaluation framework for benchmarking LLMs across vital use cases; led research assessing feasibility of hosting DataChat's custom LLMs across multiple cloud providers and platforms.
- *Expanded GenAI & ML Product Capabilities*: Spearheaded design & implementation of backend solutions; achieved scalability, reliability and compute efficiency, utilizing system design best practices and cloud native architecture.
- Consistently received high ratings across all performance evaluations.

National Science Foundation – IRIS HEP, Berkeley, CA

May 2021 - August 2021

[Fellow](#) – Graph Methods for Particle Tracking (High Luminosity Large Hadron Collider)

- *Optimized Particle Differentiation*: Developed embedding pipeline to find distance metric between 3D hit measurements pairs.
- *Accelerated Nearest Neighbors Search*: Replaced Facebook's Faiss with Fixed Radius Nearest Neighbors (FRNN) on CUDA.

Citrine Informatics, Redwood City, CA

May 2018 - August 2018

[NextGen-Fellow](#) – Computational Materials Science

- *Multi-University Research Project*: Competitively selected, successfully completed bootcamp at Stanford University, presented at NextGen Research Symposium in Golden, CO, and *co-authored a paper*.
- *Optimized Metal Defect Detection*: Trained RetinaNet and developed evaluation pipeline; achieved 85% precision & 68% recall.

EDUCATION

M.S. Computer Science, University of Wisconsin-Madison (Dec 2021)

G.P.A: 4.00/4.00

B.S. Computer Science (Minor: Mathematics), University of Wisconsin-Madison (May 2020)

G.P.A: 3.90/4.00 "Distinction in the Major" and Dean's List

LANGUAGES & TECHNOLOGY SKILLSETS

Python, Git, AWS, GCP, Azure, Terraform, Docker, Kubernetes, Helm, CircleCI, VS Code, Java, Jupyter Notebook, Unix, MacOS

LICENSES & CERTIFICATIONS

Amazon Web Services: Solutions Architect Associate; Cloud Practitioner; Machine Learning Specialty

OTHER SOFTWARE R&D PROJECTS

Autonomous Vehicles

Live Traffic-Sign Detection: Trained PyTorch MobileNet detection model; developed proof of concept for open source testbed; presented at ACM MobiArch'22.

Instance Segmentation App

Segmentation With User Images: Created app that accurately identifies and separates objects of interest from background.

Business Success & Viability Forecast

Developed Multiple ML models with Yelp Dataset: Architected feature engineering, created training and validation pipelines, and determined vital factors to predict Covid period business survival & success.

RELEVANT COURSEWORK

Machine Learning, Data Science, Artificial Intelligence, Algorithms, Computer Networks, Mobile & Wireless Networks, Computer Vision, Cryptography, Combinatorics, Spatial Web & Mobile Programming, Linear Algebra & Differential Equations, Operating Systems.

PUBLICATIONS & PRESENTATIONS

- [μCity: a miniaturized autonomous vehicle testbed](#): Procs. 17th ACM Workshop **MobiArch'22**
- [Multi defect detection & analysis of electron microscopy images with deep learning](#): Science Direct (Computational Material Sci.)
- [Graph methods for particle tracking](#): IRIS HEP fellowship final presentation